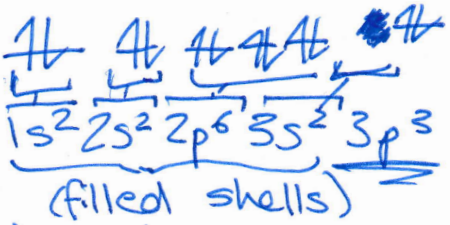
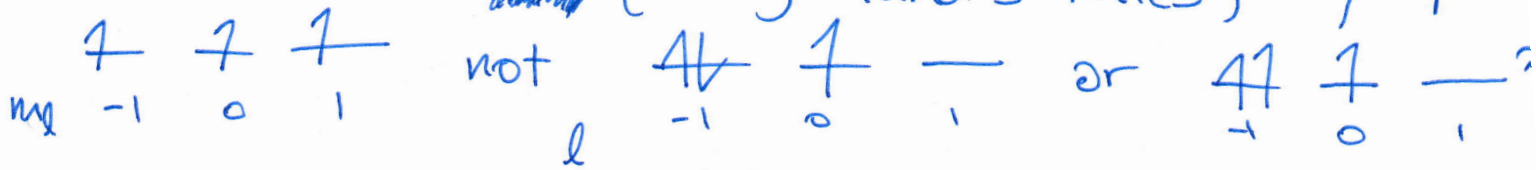
Motivating question:

P atom (ground state) electron config: $1s^2 2s^2 2p^6 3s^2 3p^3$ (filled shells)



Want to know (Using Hund's rules) why $3p^3$



Quantum number: l: $0, 1, 2 \ldots$

n: $1, 2, \ldots$ ~~$n \in n+1$~~

m_l: $-l \leq m_l \leq l$

Q: Find terms and ordering of energy levels for np^3

Hund's rules (recall)

1. For an config, the term multiplicity, has the lowest energy
2. For a multiplicity, the largest L has the lowest energy
3. For a term, if the outermost shell is half filled or less lowest J has lowest energy. If more than half filled, opposite case.

p state:

l: 1, m_l: $-1, 0, 1$

s: $1/2$, m_s: $\pm 1/2$

] $1e^-$

$M_L = \sum m_l$, $M_S = \sum m_s$] $3e^-$ ← $-3 \leq M_L \leq 3$, $-3/2 \leq M_s \leq 3/2$

$1e^-$ state combinations:

(m_l, m_s):

$(-1, -1/2), (0, -1/2), (1, -1/2)$

$(-1, 1/2), (0, 1/2), (1, 1/2)$

$M_L = 3$:

$(1, 1/2), (1, 1/2), (1, 1/2)$: $M_S = 3/2$ ↑↑↑

$(1, 1/2), (1, 1/2), (1, -1/2)$: $M_S = 1/2$ ↑↑↓

$(1, 1/2), (1, -1/2), (1, -1/2)$: $M_S = -1/2$ ↑↓↓

$(1, -1/2), (1, -1/2), (1, -1/2)$: $M_S = -3/2$ ↓↓↓

~~violates~~ violates Pauli Exclusion principle!

$M_L = 2$:

$(1, \frac{1}{2}), (1, -\frac{1}{2}), (0, \frac{1}{2}) : M_S = \frac{1}{2}$

$(1, \frac{1}{2}), (1, -\frac{1}{2}), (0, -\frac{1}{2}) : m_S = -\frac{1}{2}$

$\uparrow\downarrow$ $\uparrow$ $\uparrow$

-1 0 1

neglect $(1, -\frac{1}{2})(1, \frac{1}{2})(0, \frac{1}{2})$

$M_L = -2$:

$(-1, \frac{1}{2}), (-1, -\frac{1}{2}), (0, -\frac{1}{2}) : m_S = \frac{1}{2}$

$(-1, \frac{1}{2}), (-1, -\frac{1}{2}), (0, -\frac{1}{2}) : m_S = -\frac{1}{2}$

$\uparrow$ $\uparrow$ $-$

$\uparrow$ $\downarrow$ $-$

-1 0 1

Note that there is symmetry for configurations (up to signs for $M_L \to \pm M_L$) & M_S

$\therefore$ only consider $M_S, M_L > 0$

$M_L = 1$:

$(1, \frac{1}{2}), (-1, \frac{1}{2}), (1, -\frac{1}{2}) : M_S = \frac{1}{2}$

$(1, \frac{1}{2}), (0, \frac{1}{2}), (0, -\frac{1}{2}) : M_S = \frac{1}{2}$

$M_L = 0$:

$(1, \frac{1}{2})(0, \frac{1}{2})(-1, \frac{1}{2}) : M_S = 3/2$: $\uparrow$ $\uparrow$ $\uparrow$; -1 0 1

$(-1, \frac{1}{2})(0, \frac{1}{2})(1, -\frac{1}{2}) : M_S = \frac{1}{2}$

$(-1, \frac{1}{2})(1, \frac{1}{2})(0, -\frac{1}{2}) : M_S = \frac{1}{2}$

$(1, \frac{1}{2})(0, \frac{1}{2})(-1, -\frac{1}{2}) : M_S = \frac{1}{2}$

~~total considered~~

$M_S, M_L > 0$: 7 total considered

$\leftrightarrow$ Ground state by largest/maximal spin

$M_S = 3/2 \Rightarrow S = 3/2$, $2S+1 = 4$

$M_L = 0$, $L = 0$

Recall spectroscopic notation: ${}^{2S+1}L_J$, L: S P D F ... (0 1 2 3 ...)

$\therefore$ Ground state is: ${}^4S_{3/2}$

p^3 out of p^6: less or half filled $\therefore J = |L-S|$ is lowest in energy for Spin-Orbit / ~~S-coupling~~

$\therefore J = 3/2$

Stretched state: $M_L = L$ or $M_S = S$

if $M_L = 2 \leftrightarrow L = 2$

Whereas $M_L = 1 \leftrightarrow L = 1$ or $L = 2$ (etc.) (by $-L \leq M_L \leq L$)

$M_L = 2, M_S = 1/2$:

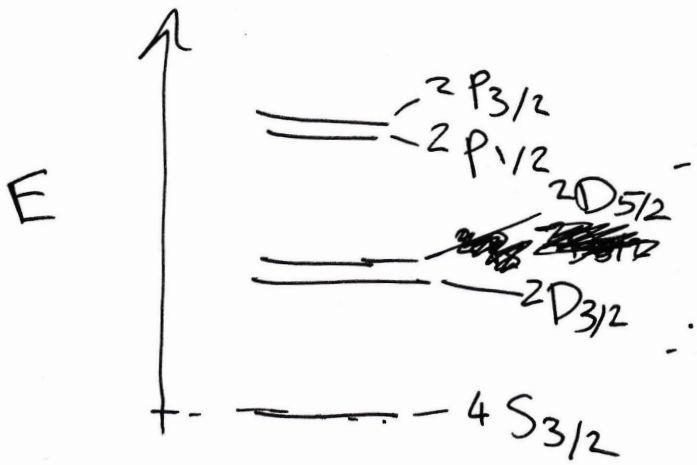
$\leftrightarrow L = 2$ ∴ D term
$S = 1/2 \Rightarrow 2S+1 = 2$
$J = |L-S|, \ldots, L+S$
$= 3/2, 5/2$
} ^2D_J ∴ $^2D_{3/2}, {}^2D_{5/2}$

$M_L = 1, M_S = 1/2$
$\leftrightarrow L = 1$ ∴ P term
$S = 1/2 \Rightarrow 2S+1 = 2$
$J = 1/2, 3/2$
} $^2P_{1/2}, {}^2P_{3/2}$

∴
terms = $^4S_{3/2}, {}^2P_{1/2}, {}^2P_{3/2}, {}^2D_{3/2}, {}^2D_{5/2}$

Before: ($M_S, M_L > 0$) 7 total states
↳ for all M_S, M_L is 20 total states

Recall: multiplicity $(2J+1) = (2S+1)(2L+1)$
↳ above terms also give 20 total states



^2P or ^2D
($L=1$) ($L=2$)

∴ (Hund's rule 2:)
$E(^2D_{3/2}) < E(^2P_{3/2})$

∴ (Hund's rule 3):
$E(^2D_{3/2}) < E(^2D_{5/2})$
($J = 3/2$) ($J = 5/2$)